POLITEKNIK MALAYSIA
TUN SYED NASIR



CHAPTER 1: INTRODUCTION TO WASTE MANAGEMENT IN OIL AND FAT INDUSTRY | Ts. MUHAMMAD AZRI IZANI BIN MOHAMAD HALIM

INTRODUCTION

- Oils and fats are the most abundance lipid in nature, consists of mixture of saturated and unsaturated fatty acids.
- Structurally, both are esters of glycerol with three fatty acids, called either triacylglycerols or triglycerides.

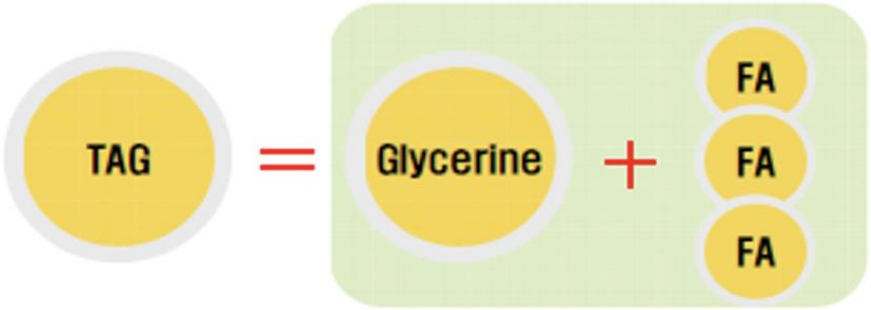
INTRODUCTION

- Oil and fats comparison:

Characteristics	Oils	Fats
Fatty Acid Composition	Contain more unsaturated fatty acids	Contain more saturated fatty acids
Physical State at Room Temperature	Usually liquid	Usually solid
Melting Point	Lower melting point	Higher melting point
Source	Mainly from plants	Mainly from animals and some tropical plants
Examples	Soybean oil, Sunflower oil, Olive oil	Animal fats, Palm oil, Coconut oil

INTRODUCTION

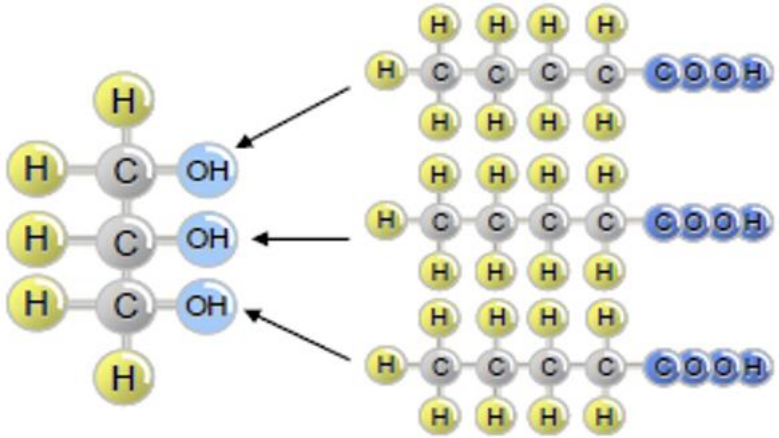
- **Triacylglycerols (TAG)**, also known as a **triglyceride**, is the main form of fat and oil found in plants and animals.



Esterification process

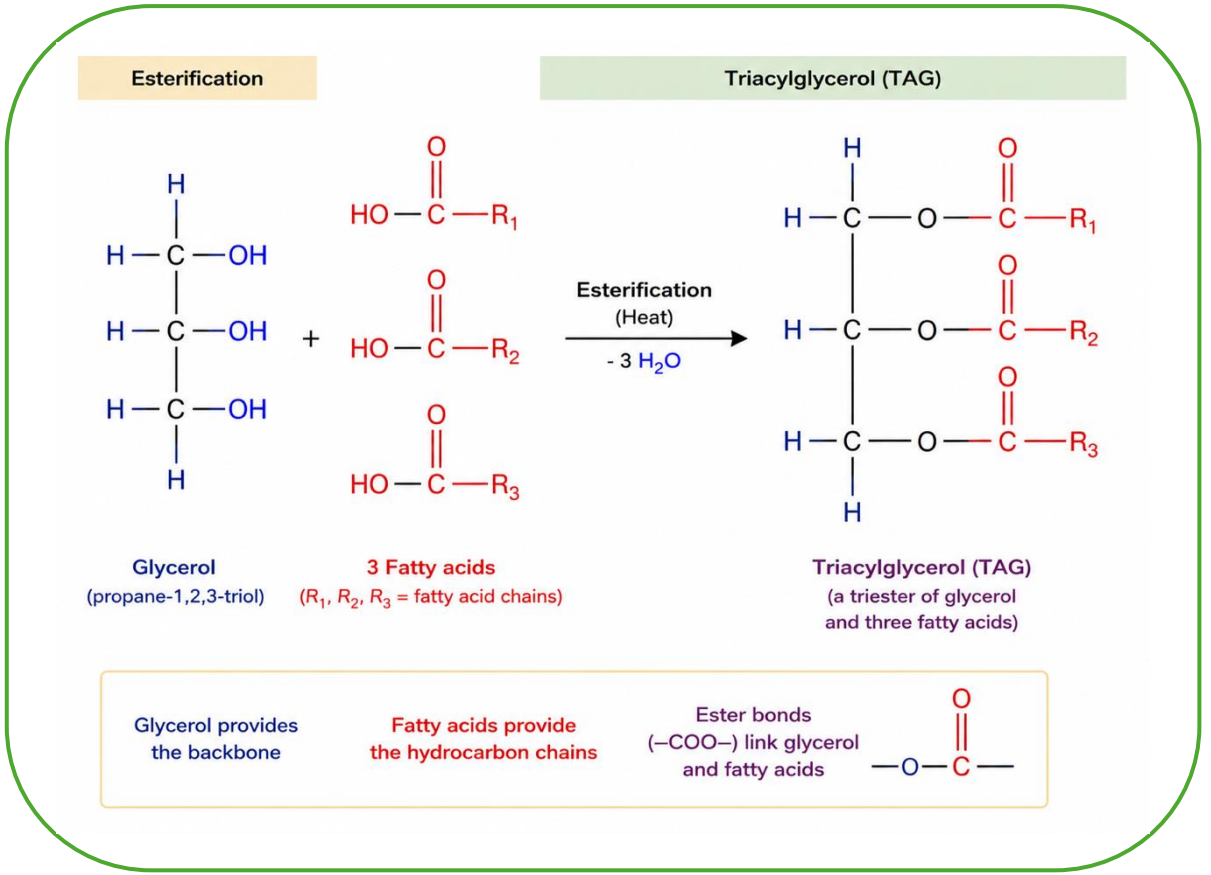
INTRODUCTION

- **Esterification process**



Glycerol

Fatty Acid



INTRODUCTION

- **Classification of oils & fats:**

Vegetables

Vegetables oils:
- Soybeans oil
- Sesame oil
- Olive oil
- Sunflower oil
- Peanut oil

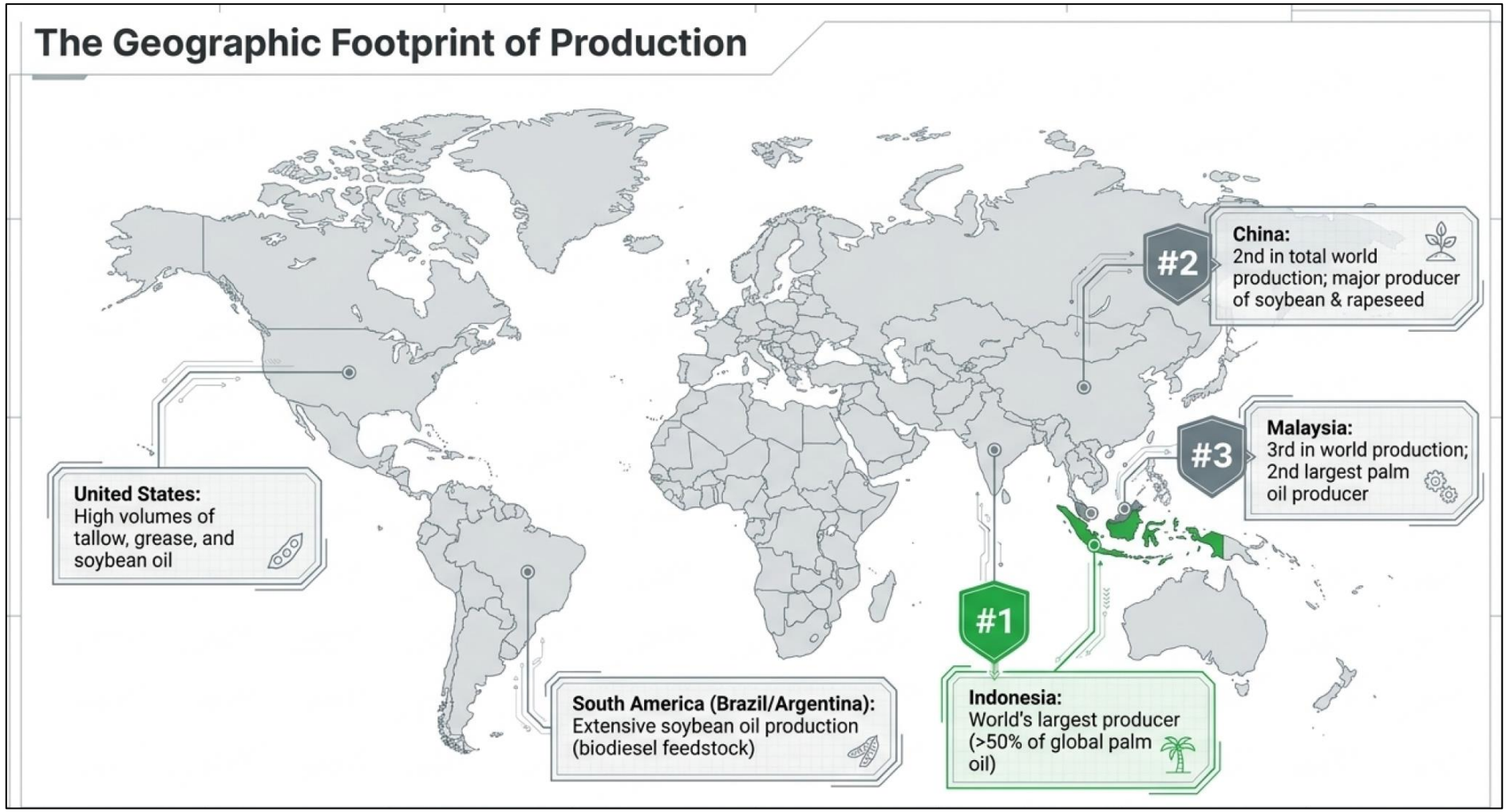
Vegetables fats:
- Coconut oil
- Palm oil
- Palm kernel oil
- Cacao fat

Animals & marines

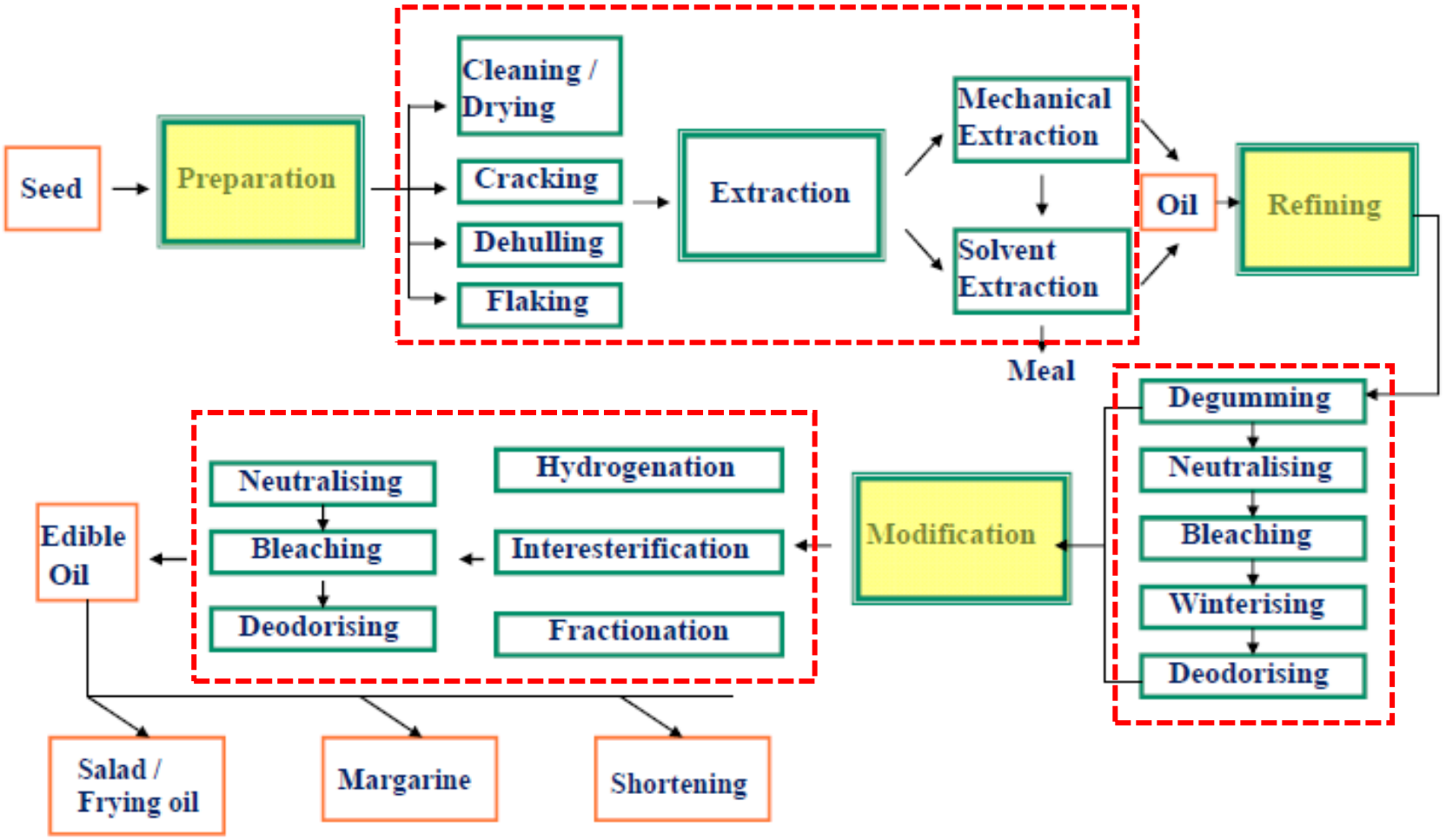
Marine oil:
- Fish oil

Animal fats:
- Tallows
- Lards
- Butter

INTRODUCTION



GENERAL OIL & FAT PROCESSING FLOW



EXAMPLE 1

1. State 3 main stages in oil and fats processing flow.
2. State 5 processes in refining stage

TYPE OF WASTE*

- Sources of waste from the oil and fat industry come from different stages such as plantation (Oil plantation), primary processing (Milling & Extraction) and secondary processing (Refining & Product Manufacturing). All these stages produce different type of wastes (solid waste, wastewater and air emissions).

- Plantation main type of waste:

1. Biomass from deforestation (Empty fruit bunches (EFB) *(especially oil palm)*, Fronds and pruned leaves, Trunk waste (old oil palm trees during replanting), Fruit bunch residues (unharvested or damaged fruits)
2. Air emission from open burning
3. Agricultural runoff water containing fertilizers and pesticides
4. Soil residues mixed with organic matter

TYPE OF WASTE

- Primary processing main type of waste:

1. Palm Oil Mill Effluent (POME) / process wastewater
2. Palm kernel shell
3. Mesocarp fibre
4. Oil sludge (semi-liquid mixture)
5. Oil-contaminated wastewater
6. Spent cleaning water from sterilization and clarification

TYPE OF WASTE

- Secondary processing main type of waste:

1. Spent bleaching earth (SBE)
2. Soapstock (from neutralization process)
3. Acid oil (from soapstock splitting)
4. Deodorizer distillate (contains FFA, sterols, tocopherols)
5. Used cooking oil (from food industry frying operations)
6. Glycerol (crude glycerin from biodiesel production)
7. Chemical-laden wastewater (alkali, acids, salts)

EXAMPLE 2

1. Explain with example of 3 main type of oils and fats industry waste.

SOURCE OF WASTE

Source of oils and fats waste

- **Solid waste**
- **Wastewater**
- **Air Emissions**

1. **Bleaching process** (Spent bleaching clay remain as waste after this clay used in edible oil refining to remove impurities and improve oil colour, quality, and stability)
2. **Hydrogenation process** (Spent hydrogenation nickel catalyst remains as waste after the hydrogenation of oils and fats)
3. **Plantation process** (Solid materials generated during the processing of oil crops or animal fats)
4. **Oil extraction process** (Oilseed meal or oilseed cake is the solid material left behind after oil has been extracted from the seeds)

SOURCE OF WASTE*

Source of oils and fats waste

- **Solid waste**
- **Wastewater**
- **Air Emissions**

1. **Oilseed plantation process** (harvested oilseeds and farming equipment are washed, the dirty water carries soil, plant residues, and agricultural chemicals, producing wastewater)
2. **Oilseed preparation & Extraction process** (seed cleaning and oil extraction, water becomes contaminated with oil droplets and seed residues, forming wastewater)
3. **Crude oil refining process** (Degumming, neutralization and washing process generate wastewater contaminated with free fatty acid, oil and chemical)
4. **Refined oil modification process** (Equipment cleaning and cooling operations generate wastewater)

SOURCE OF WASTE

Source of oils and fats waste

- **Solid waste**
- **Wastewater**
- **Air Emissions**

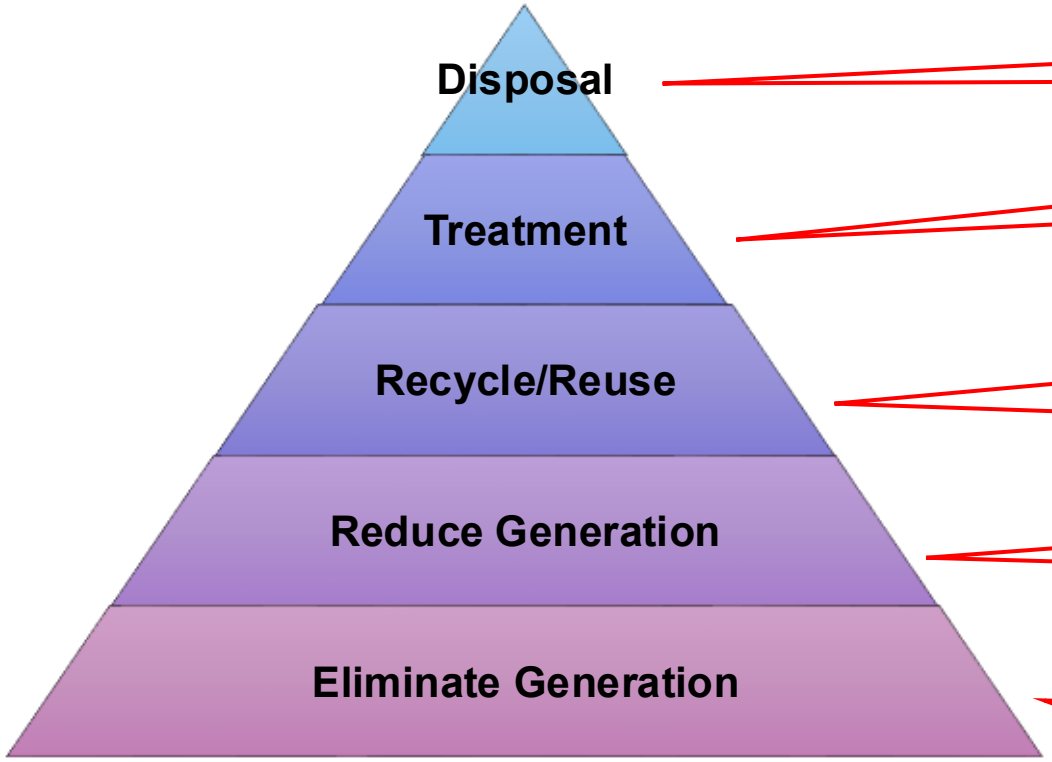
1. **Oilseed preparation & handling** (When dry oilseeds are moved, crushed, or processed, small dust particulate form)
2. **Solvent extraction process** (solvent used to extract oil evaporates and escapes into the air, producing solvent emissions)
3. **Oil deodorization process** (Strong-smelling compounds are removed from oil during refining and released as odor emissions)
4. **Steam generation process in boiler** (When fuel is burned to produce steam, smoke and gases are released through the boiler chimney)

WASTE MANAGEMENT

- Waste management: Application of techniques to ensure an efficient execution of the various functions of collection, transportation, processing, treatment, recovery, and disposal of solid, liquid, and gaseous wastes generated from domestic, commercial, agricultural, and industrial activities.
- Various waste management approaches can be implemented depending on the type and characteristics of the waste.
- The fundamental principles of waste management are to minimize and recycle the waste, recover the energy and finally dispose the waste.

WASTE MANAGEMENT HIERARCHY PRINCIPLE

- Waste management is systematic process of reducing, collecting, treating, recycling, recovering, and safely disposing waste sustainably.



Last waste management option

Reduction of volume and toxicity level of hazardous waste

Maximize recycling and reuse of hazardous substances such as solvents and acids before treatment or disposal is required.

Minimizing the amount of waste production.

Eliminate hazardous chemicals by modifying processes or substituting materials that avoid hazardous waste generation.

EXAMPLE 3

1. Explain 3 principles of waste management.
2. State 3 wastewater source from oil and fats industry.

WASTE DISPOSAL IMPACT

Source / Process	Impact	Explanation
Plantation (open burning)	Global warming	Burning biomass releases CO_2 into the air.
Plantation residues	Soil contamination	Oil leaks from husks, fibres, and shells pollute soil.
Boiler operations	Air pollution	Smoke and gases cause dirty air.
Sterilization & cooling water	Wastewater generation	Dirty water is produced when mixed with waste.
POME (Palm Oil Mill Effluent)	Water pollution	High organic waste reduces water quality and oxygen.
POME (anaerobic condition)	Greenhouse gas emission	Methane gas is released, contributing to climate change during anaerobic decomposition.

WASTE DISPOSAL IMPACT

Process	Impact	Explanation
Deodorization	Air pollution (odor emissions)	Smelly volatile compounds are released into the air.
Bleaching process	Soil & water contamination	Spent bleaching earth can leak oil into soil and groundwater.
Hydrogenation process	Soil & water pollution	Used nickel catalyst can contaminate soil and water.
Soapstock acidulation	Water pollution & odor	Sulphate-rich wastewater increases odor and total dissolved solids (TDS).

IMPORTANCE OF WASTE MANAGEMENT*

These are the importances of waste management system:

1. **Conservation of Natural Resources:** It promotes the efficient use of materials and supports recycling and reuse, thereby reducing the demand for virgin resources and protecting natural ecosystems.
2. **Environmental Protection:** Proper waste management minimizes the risk of severe environmental damage caused by improper handling, treatment, or disposal of waste.
3. **Reduction of Environmental Pollution:** It helps reduce air, water, and soil pollution, including the emission of greenhouse gases such as methane and carbon monoxide from unmanaged waste.
4. **Cost Efficiency:** An effective industrial waste management system can lower operational and production costs through material recovery, energy savings, and reduced waste disposal expenses.
5. **Employment Opportunities:** The development and operation of waste management systems create new job opportunities in areas such as collection, treatment, recycling, and environmental monitoring.

THANK YOU

PLEASE SUBMIT YOUR MIND MAP